

AN AI POWERED SOFTWARE-BASED FIRE DETECTION SYSTEM FOR CCTV PLATFORMS.

FiremeX - TEST PLAN

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Project	FiremeX
Prepared By	Jayashini Jayaweera
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1. Objectives

- Verify fire and smoke detection accuracy under varied lighting, weather, and camera conditions.
- Validate real-time alert generation and notification delivery within defined SLA (≤ 5 seconds).
- Confirm integration with camera feeds (ONVIF/RTSP), dashboard UI.
- Ensure system reliability for 24x7 continuous monitoring with minimal false positives/negatives.
- Validate data storage, audit logging, and historical incident retrieval.
- The system uses historical alerts and incident records to identify recurring patterns and continuously improve the AI model, reducing false alarms and increasing detection accuracy for frequently occurring scenarios.

2. Scope

2.1 In Scope

- Functional testing of video ingestion, detection engine, alerting, and dashboard modules.
- Non-functional testing: performance, load, security, and reliability of the detection pipeline.
- Integration testing with CCTV/NVR systems, notification gateways (SMS/Call).
- Regression testing for every AI model version update.
- Usability testing of the monitoring dashboard for control-room operators.

2.2 Out of Scope

- Testing of third-party fire-suppression hardware actuation (mechanical).
- Detailed AI model training pipeline validation (covered under Data Science validation plan).

3. Test Approach / Methodology

Test Level	Description	Owner
Unit Testing	Developer-level testing of detection algorithm modules and APIs	QA Members
Integration Testing	Camera feed ingestion, AI engine, alert & notification module integration	QA Members
System Testing	End-to-end validation of fire detection to alert delivery flow	QA Members
Performance Testing	Multi-camera concurrent stream load and detection latency testing	QA Members
Security Testing	Access control, API vulnerability testing	QA Members
UAT	Control room operators validate real-world usability and alert clarity	End Users
Regression Testing	Re-validation after every model & software build update	QA Members

4. Test Environment

- Staging environment mirroring production: IP CCTV cameras (indoor/outdoor), NVR, GPU inference server.
- Simulated fire/smoke video datasets (day, night, fog, rain) for regression suites.
- Test dashboard instance connected to a sandbox SMS/Call gateway and mock fire-panel API.

5. Entry Criteria

- Build deployed successfully to QA/staging environment.
- Unit testing completed with $\geq 90\%$ code coverage on core detection modules.
- Test data sets (fire/smoke/no-fire video clips) prepared and validated.
- Test cases reviewed and approved.

6. Exit Criteria

- 100% of planned test cases executed; $\geq 95\%$ pass rate.
- No open Critical or High severity defects.
- Detection accuracy $\geq$ agreed threshold (e.g., 95% true positive, $\leq 2\%$ false positive on validation set).
- Performance and security sign-off obtained.

7. Suspension & Resumption Criteria

Testing will be suspended if the detection engine fails to start, camera feed ingestion is completely broken, or a Critical defect blocks more than 30% of planned test cases. Testing resumes once a fix is deployed and smoke-tested.

8. Roles and Responsibilities

Role	Responsibility
QA	Test planning, strategy, sign-off, stakeholder reporting
	Test case design, execution, defect logging
	Regression automation suite for detection & UI
	Load/stress testing of video pipeline
Business Analyst	Requirement clarification, UAT coordination
Developers	Defect fixes, unit testing, environment support

9. Test Deliverables

- Test Plan, Test Cases, Traceability Matrix
- Test Execution Report, Defect Reports
- Performance & Security Test Reports
- Test Closure Report

10. Schedule

Phase	Start	End
Test Planning & Design	Week 06	Week 09
Test Environment Setup	Week 10	Week 11
Functional Testing	Week 12	Week 21
Integration Testing	Week 18	Week 21
Performance & Security Testing	Week 21	Week 24
UAT	Week 25	Week 25
Test Closure & Sign-off	Week 27	Week 28

11. Risks and Mitigation

Risk	Mitigation
Insufficient real fire/smoke video samples for testing	Use synthetic/augmented datasets and controlled test burns from vendor labs
High false-positive rate impacting reliability	Threshold tuning cycles with QA feedback before each release
Network latency affecting real-time alerting	Dedicated performance test on production-like network conditions
Camera hardware variability (brand/resolution)	Test matrix covering multiple camera makes/models supported